STUDENT CODE: **100** Question Type: **A**

Gebze Technical University

Department of Computer Engineering
CSE 312 Spring 2026 Quiz 8 Apr 22 2026

PART A: Answer on the front side:
Consider a system with two processes (P1, P2) and two resources (R1, R2):
- P1 holds R1 and is waiting for R2
- P2 holds R2 and is waiting for R1

a) Does a deadlock occur? Justify your answer. (2 pts)
b) List the four Coffman conditions required for deadlock. (4 pts)
c) Suggest one method to prevent deadlock and explain briefly. (4 pts)

PART B: Answer on the backside:

STUDENT CODE: **100** Question Type: **A**

A system has **3 page frames**. The page reference string is:

1 2 3 4 1 2 5 1 2 3 4 5

a) Using **FIFO**, fill the frame table and find the total number of page faults. (4 pts)

b) Using **LRU**, fill the frame table and find the total number of page faults. (4 pts)

c) Which algorithm performs better for this reference string? Briefly explain. (2 pts)